



Introduction to Bioengineering Theory (BBL1020; 3-0-0)

Lecture-14: 1 Class

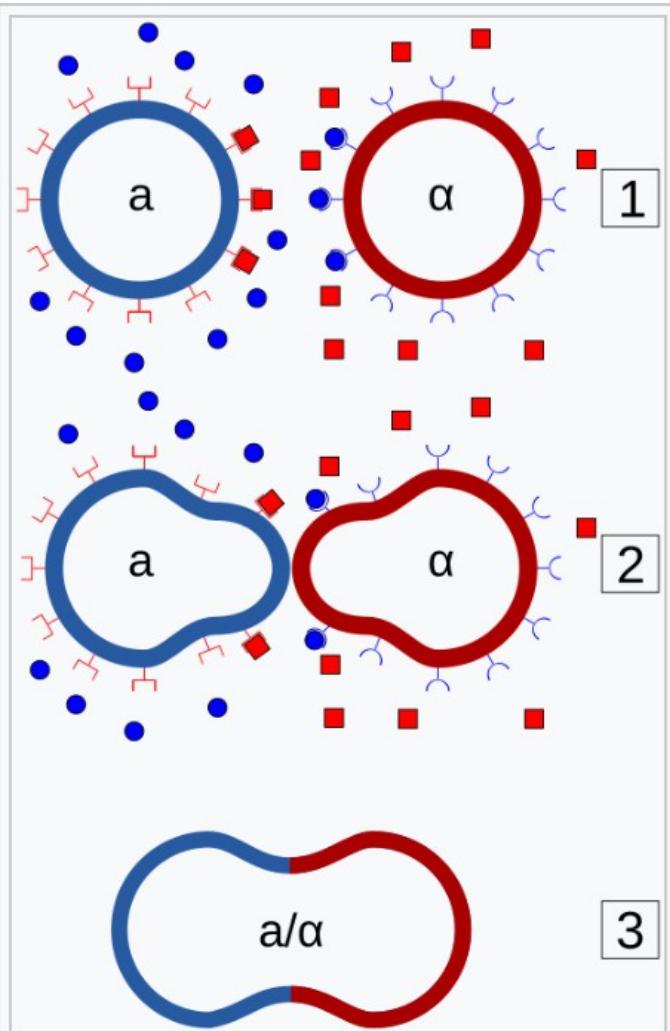


Please listen Carefully

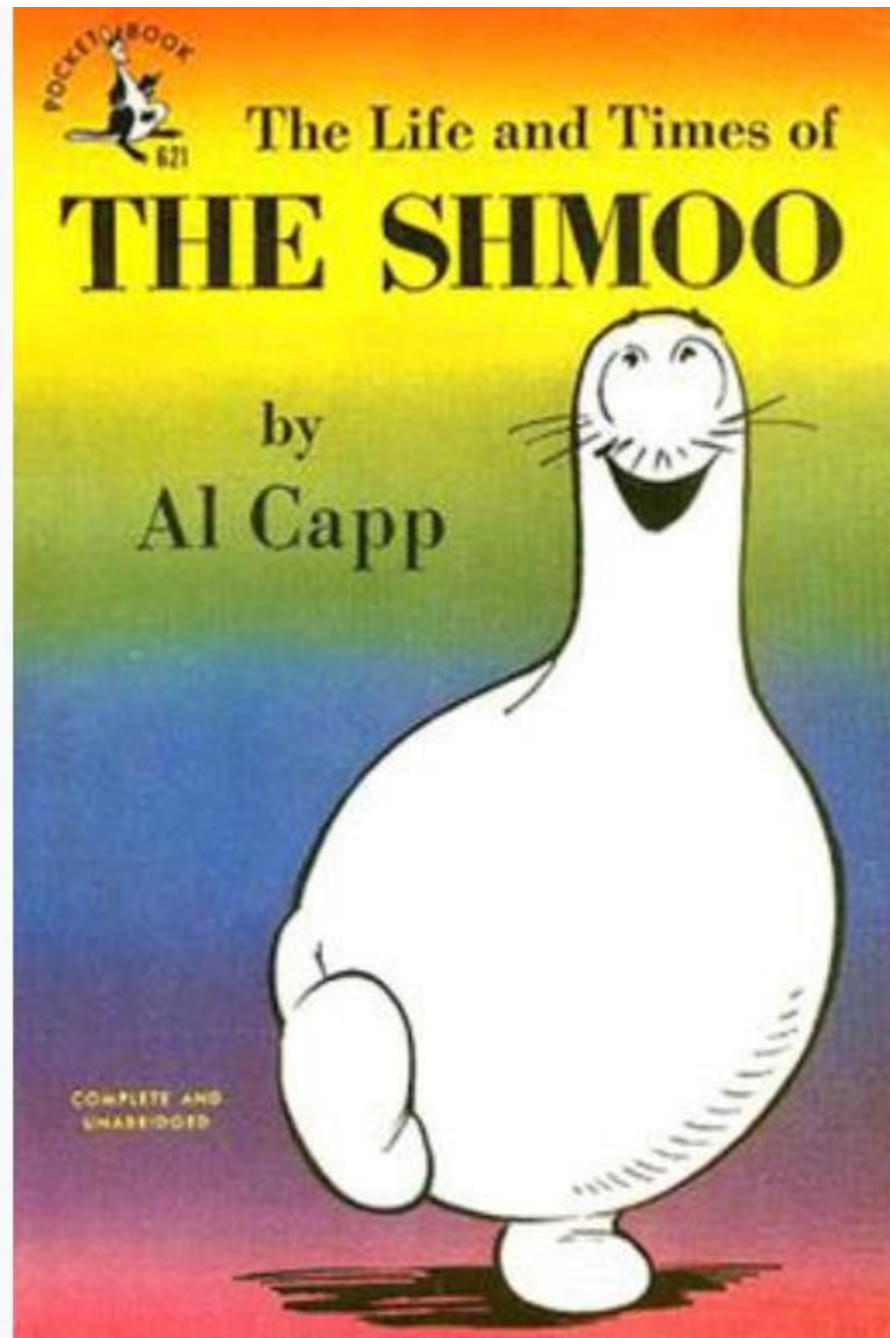
&

Keep your phone(s) on silent mode

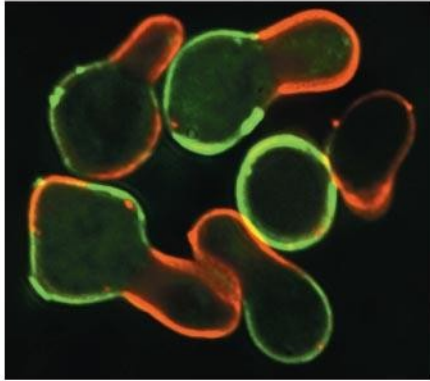
-
- Signaling pathways can also affect the physical characteristics of a cell, for example, cell shape



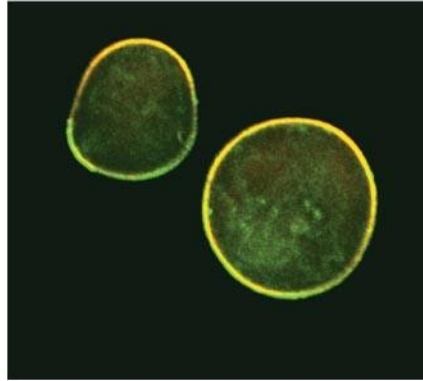
Two haploid yeast of opposite mating types secrete pheromones, grow projections, and mate



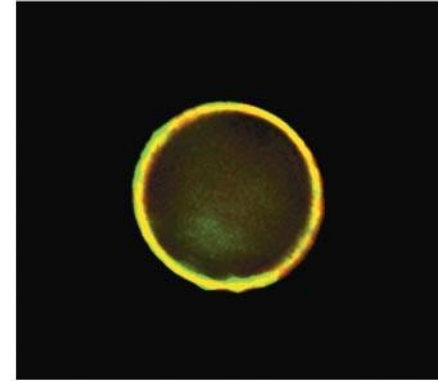
RESULTS



Wild-type (shmoos)

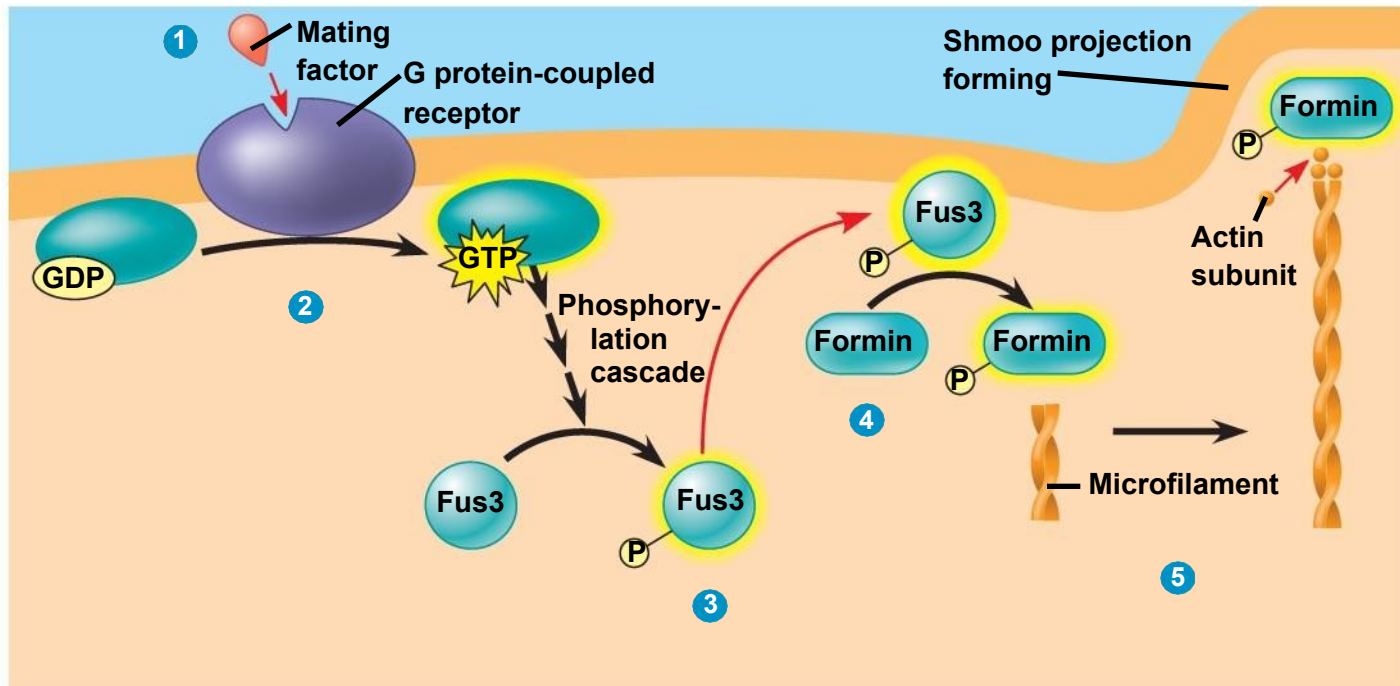


Δ Fus3



Δ formin

CONCLUSION



Fine-Tuning of the Response

- Multistep pathways have two important benefits:
 - Amplifying the signal (and thus the response)
 - Contributing to the specificity of the response

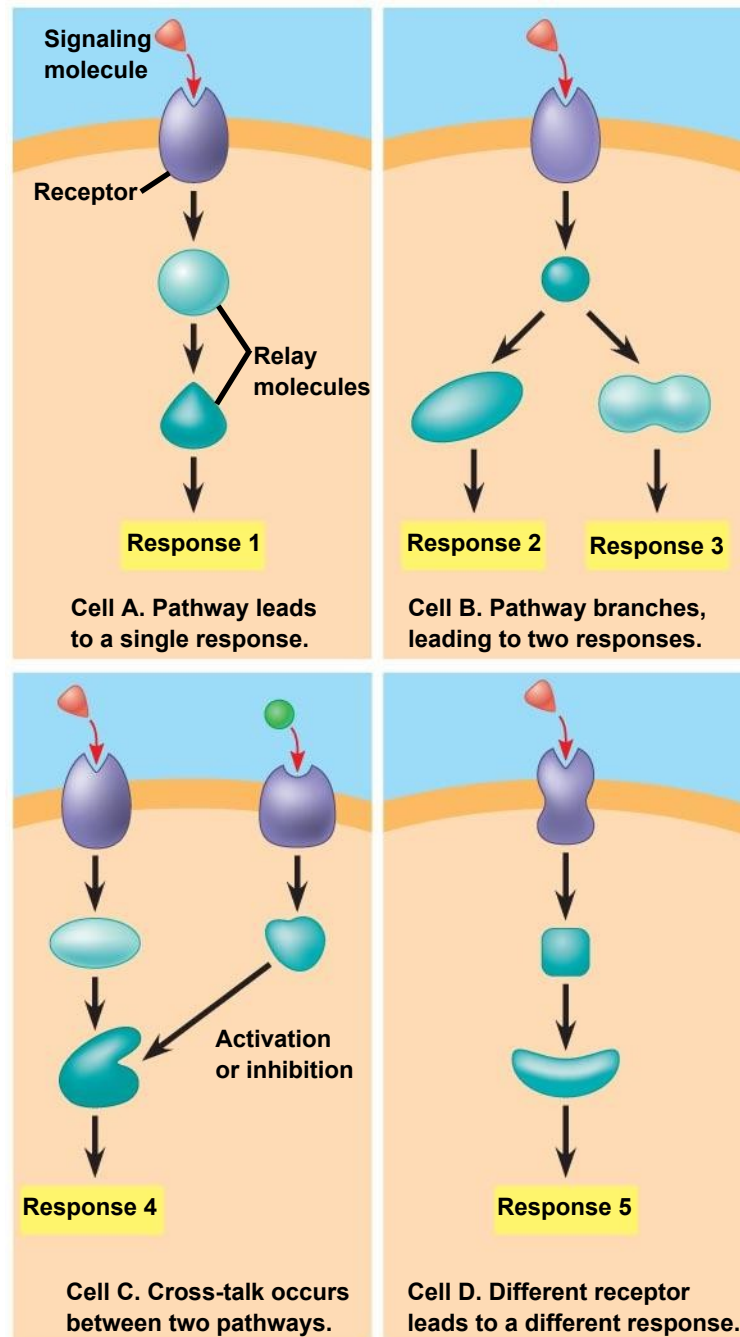
Signal Amplification

- Enzyme cascades amplify the cell's response
- At each step, the number of activated products is much greater than in the preceding step

The Specificity of Cell Signaling and Coordination of the Response

- Different kinds of cells have different collections of proteins
- These different proteins allow cells to detect and respond to different signals
- Even the same signal can have different effects in cells with different proteins and pathways
- Pathway branching and “cross-talk” further help the cell coordinate incoming signals

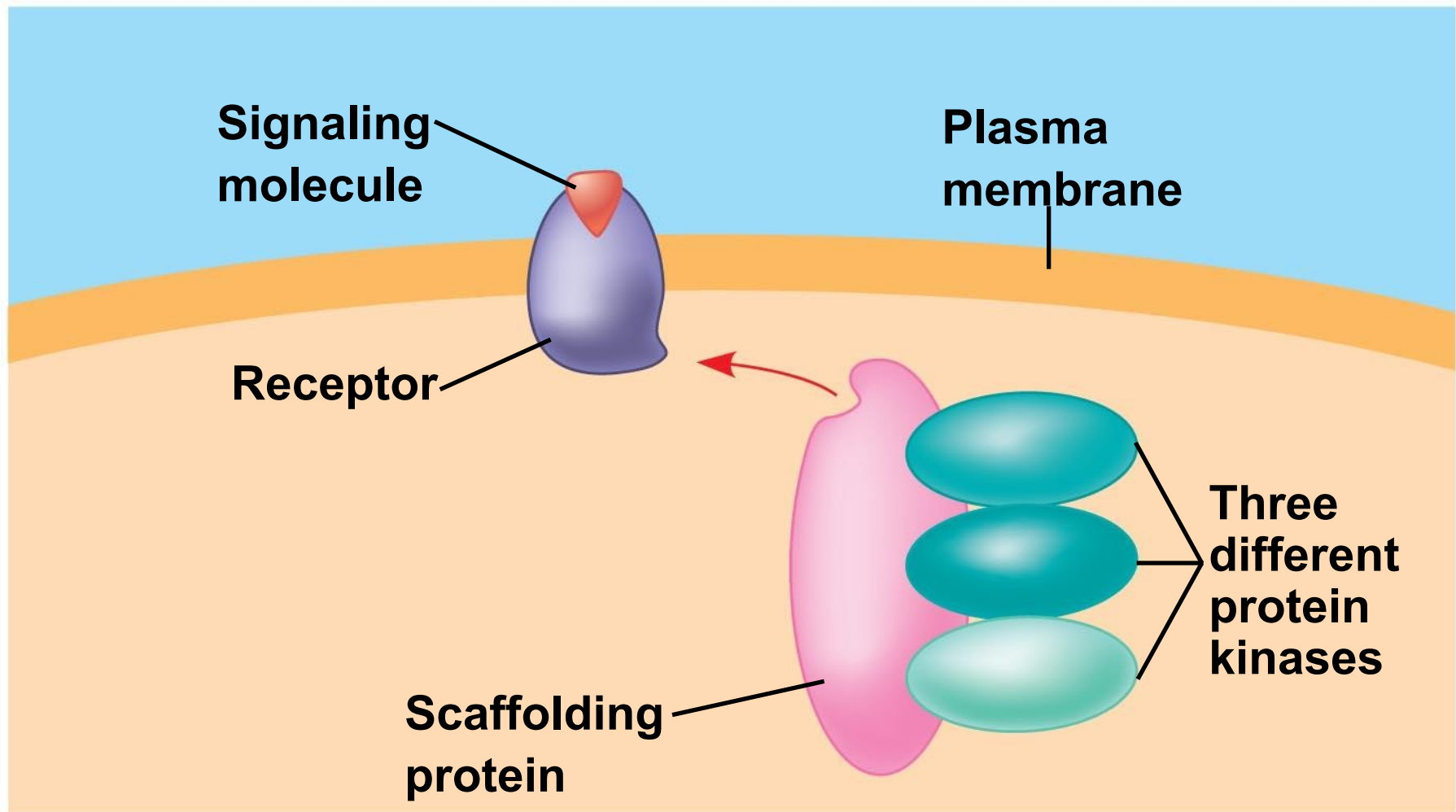
Fig. 11-17



Signaling Efficiency: Scaffolding Proteins and Signaling Complexes

- **Scaffolding proteins** are large relay proteins to which other relay proteins are attached
- Scaffolding proteins can **increase the signal transduction efficiency by grouping together different proteins involved in the same pathway**

Fig. 11-18



Termination of the Signal

- Inactivation mechanisms are an essential aspect of cell signaling
- When signal molecules leave the receptor, the receptor reverts to its inactive state

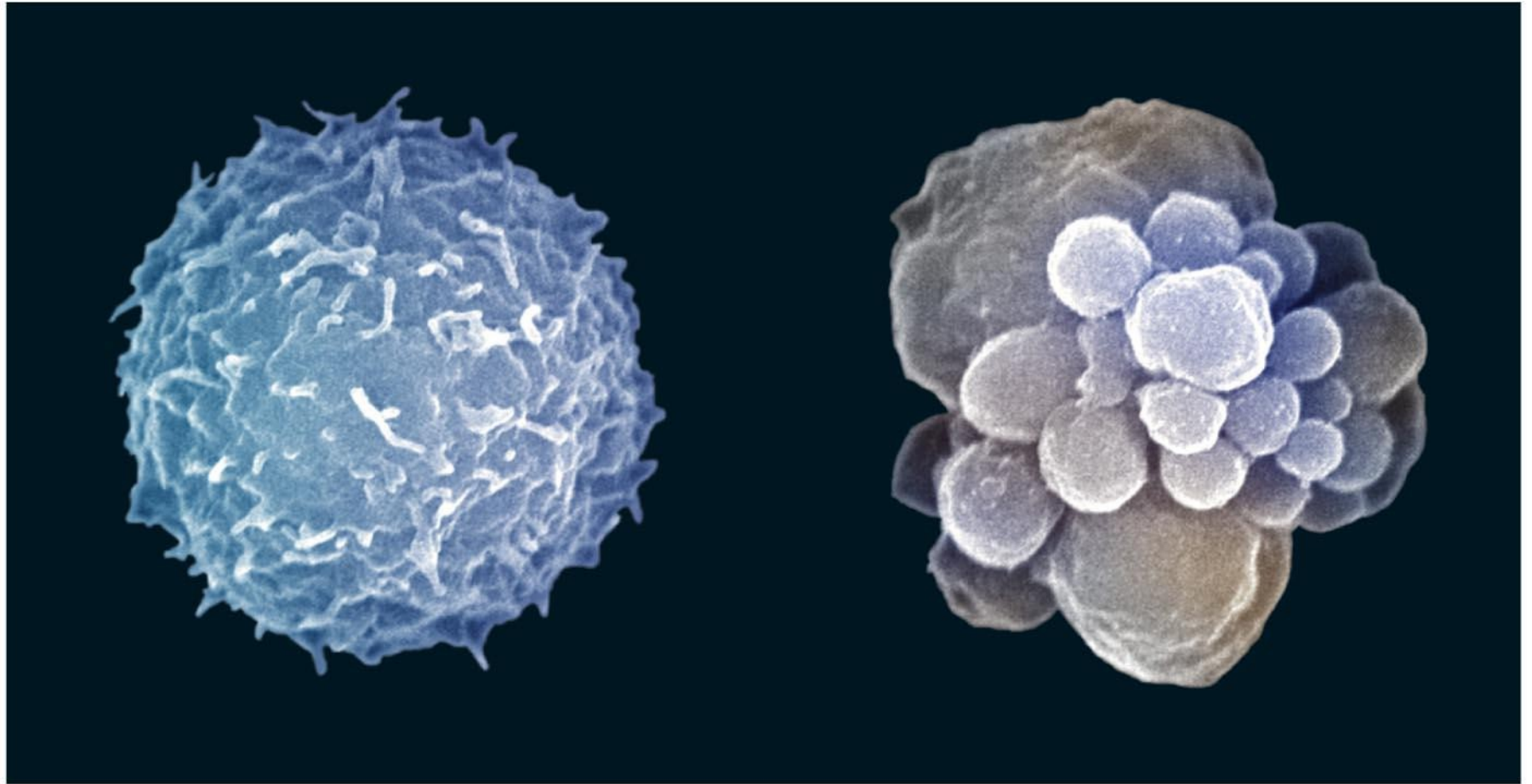
Concept 11.5: Apoptosis (programmed cell death) integrates multiple cell-signaling

pathways

Apoptosis is programmed or controlled cell suicide

- A cell is chopped and packaged into vesicles that are digested by scavenger cells
- Apoptosis prevents enzymes from leaking out of a dying cell and damaging neighboring cells

Fig. 11-19

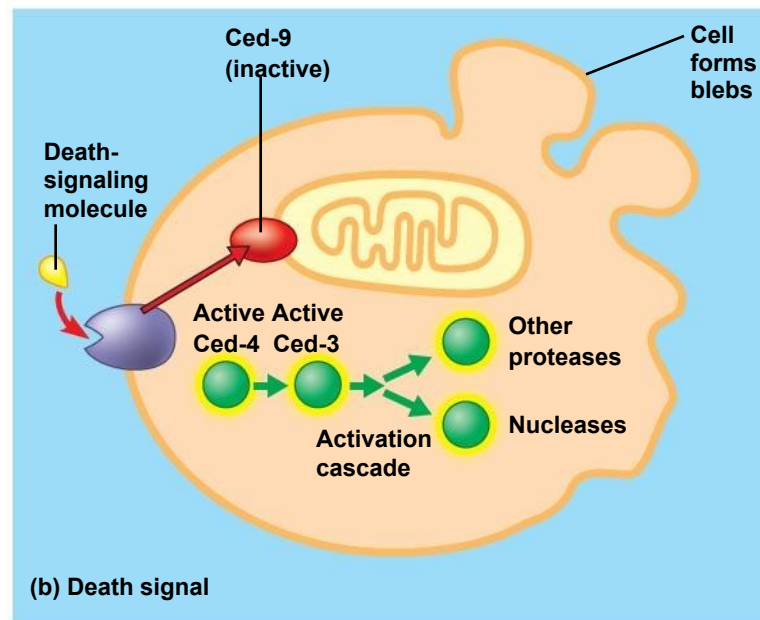
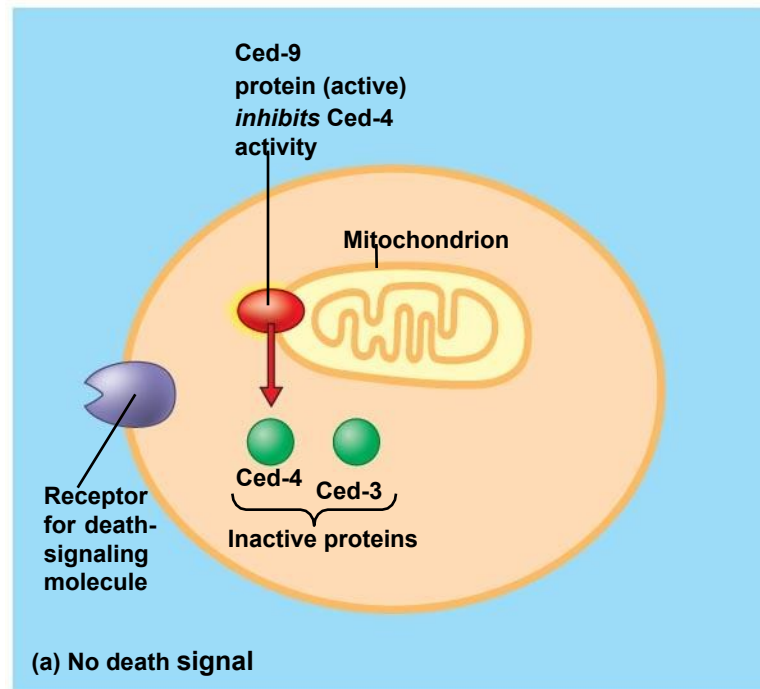


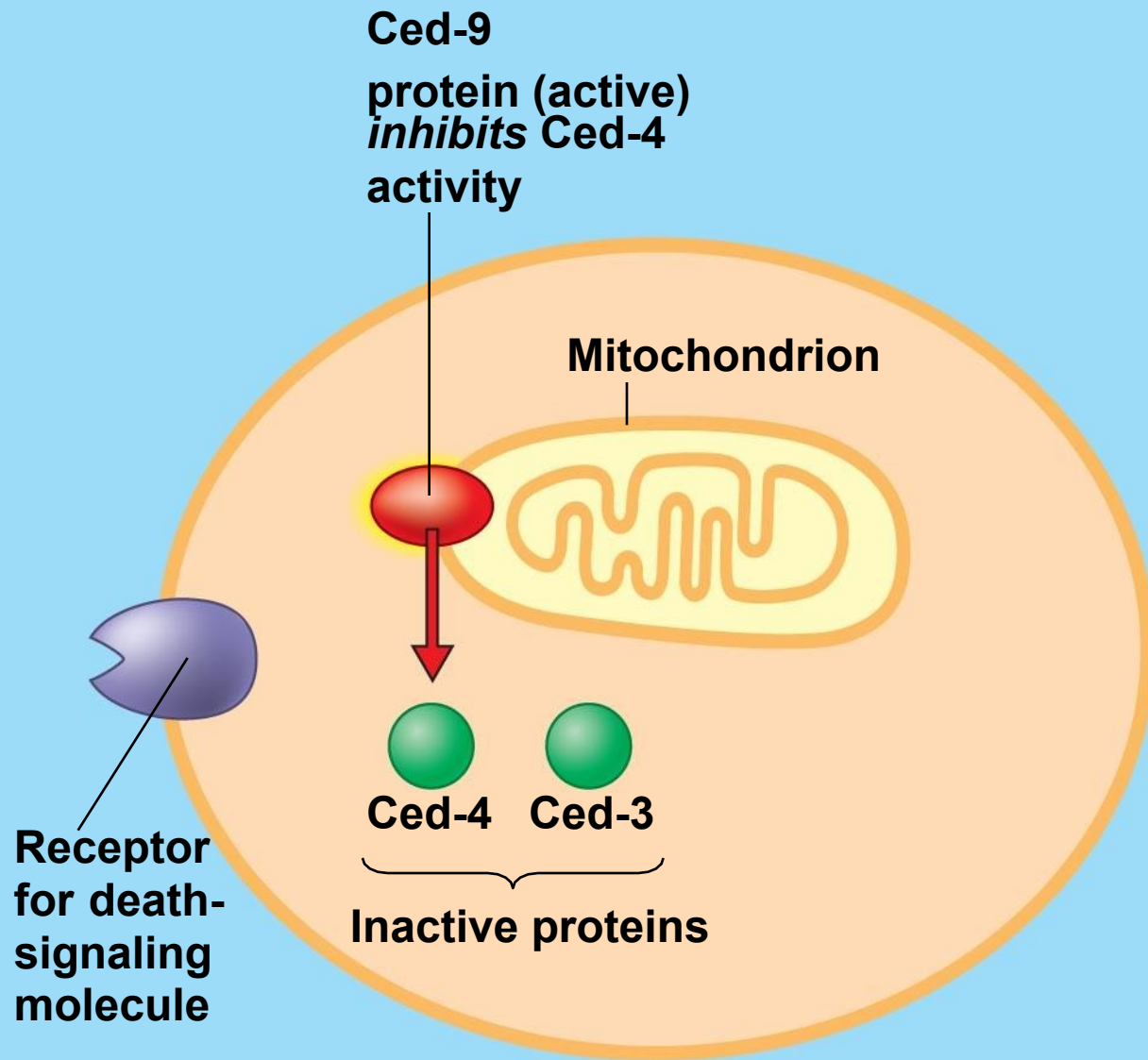
2 μm

Apoptosis in the Soil Worm *Caenorhabditis elegans*

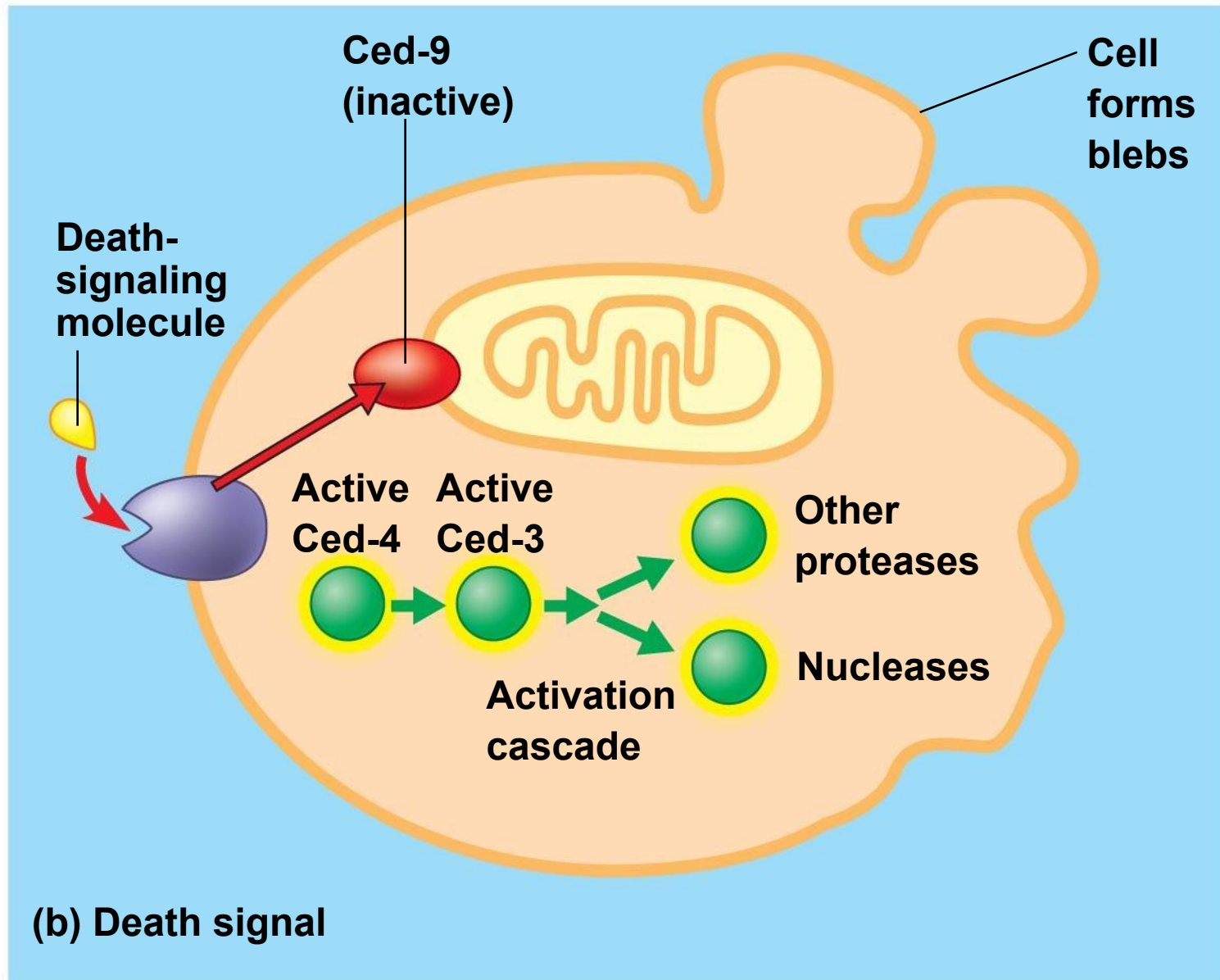
- Apoptosis is important in shaping an organism during embryonic development
- The role of apoptosis in embryonic development was first studied in *Caenorhabditis elegans*
- In *C. elegans*, apoptosis results when specific proteins that “accelerate” apoptosis override those that “put the brakes” on apoptosis

Fig. 11-20





(a) No death signal



Apoptotic Pathways and the Signals That Trigger Them

- Caspases are the main proteases (enzymes that cut up proteins) that carry out apoptosis
- Apoptosis can be triggered by:
 - An extracellular death-signaling ligand
 - DNA damage in the nucleus
 - Protein misfolding in the endoplasmic reticulum

-
- Apoptosis evolved early in animal evolution and is essential for the development and maintenance of all animals
 - Apoptosis may be involved in some diseases (for example, Parkinson's and Alzheimer's); interference with apoptosis may contribute to some cancers

Interdigital tissue

